

Equação da Quantidade de Movimento

Sabemos a *equação da quantidade de movimento na sua forma integral*.

$$\vec{F} = \vec{F}_s + \vec{F}_B = \frac{\partial}{\partial t} \int_{VC} \vec{V} \rho dV + \int_{SC} \vec{V} \rho \vec{V} d\vec{A}$$

Na forma diferencial expressamos as equações para um sistema infinitesimal de massa dm , para a qual a segunda lei de Newton pode ser expressa como:

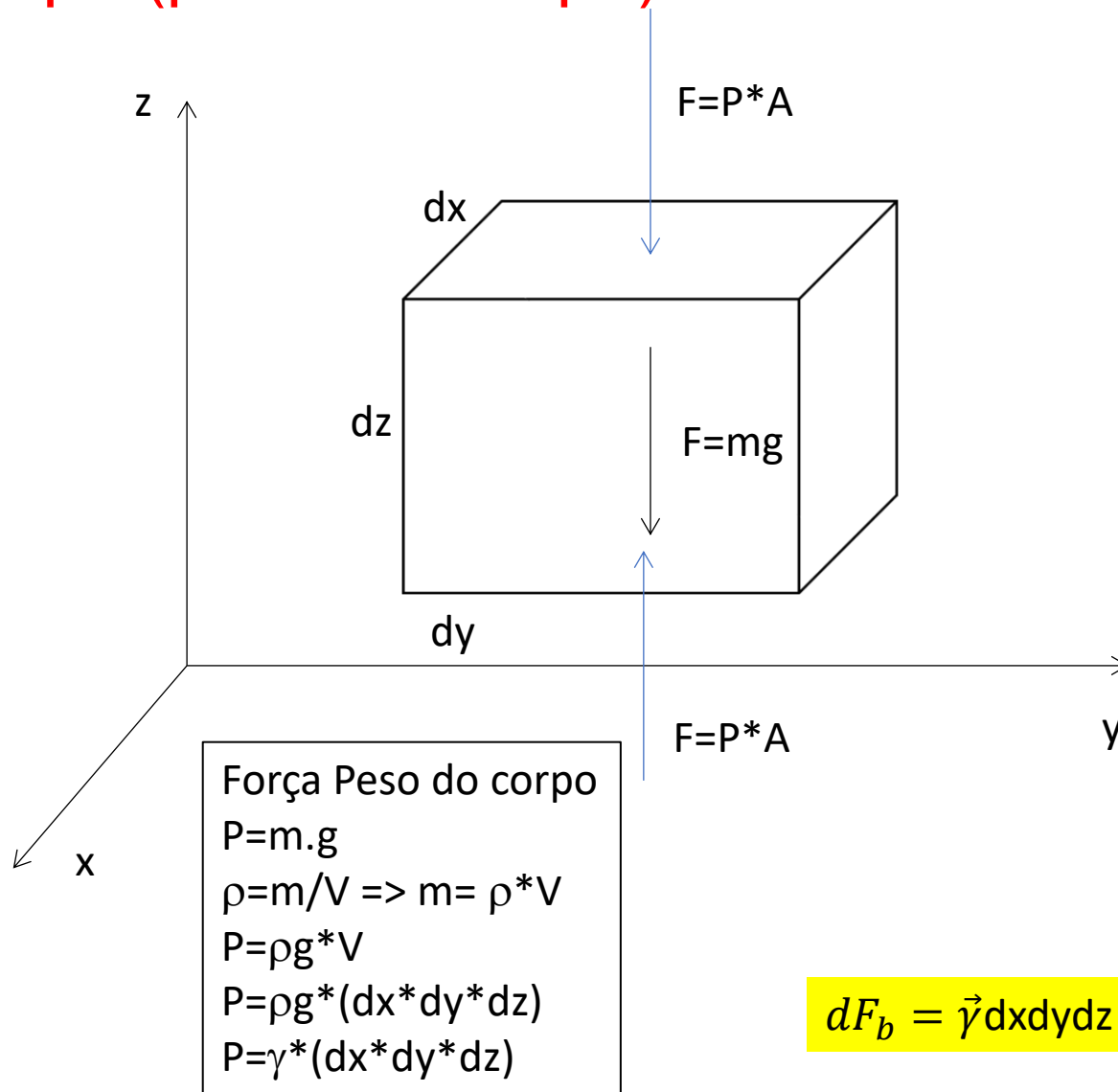
$$d\vec{F} = dm \frac{d\vec{V}}{dt} \Bigg)_{\text{sistema}}$$

o termo dm é facilmente determinado pelo produto entre a massa específica do fluido dentro do V.C. e o volume diferencial.

$$\frac{d\vec{V}}{dt} \Bigg)_{\text{sistema}} = \frac{D\vec{V}}{Dt}$$

Equação da Quantidade de Movimento

Força Campo (peso do Corpo)



$$F = \text{Força Peso} = P$$

$$m \frac{d\vec{V}}{dt} = \rho g \cdot (dx \cdot dy \cdot dz)$$

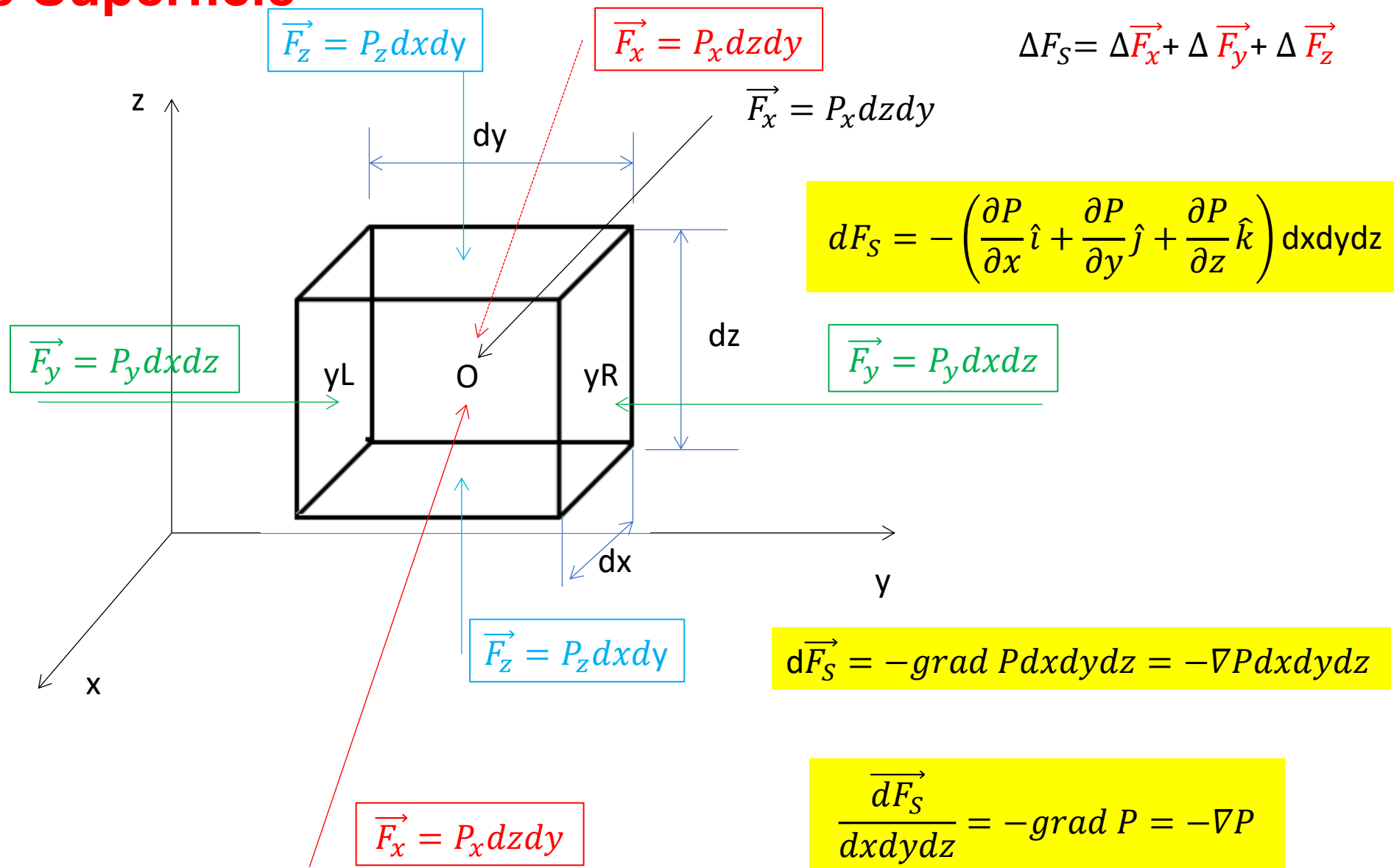
$$\rho dx dy dz \frac{d\vec{V}}{dt} = \rho g \cdot (dx \cdot dy \cdot dz)$$

$$\frac{d\vec{V}}{dt} = \vec{g}$$

$$dF_b = \vec{\gamma} dx dy dz$$

Equação da Quantidade de Movimento

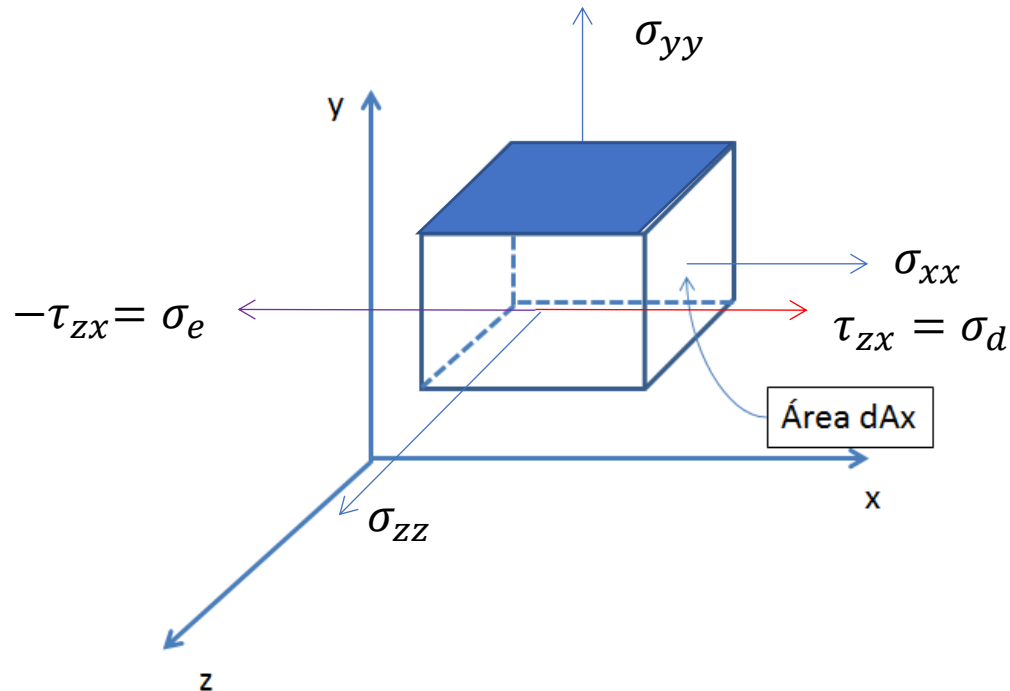
Força de Superfície



FÍSICA DOS FLUÍDOS

Tensões normais e tangenciais num elemento de fluido

Tensões de Superfície num Elemento de Fluido

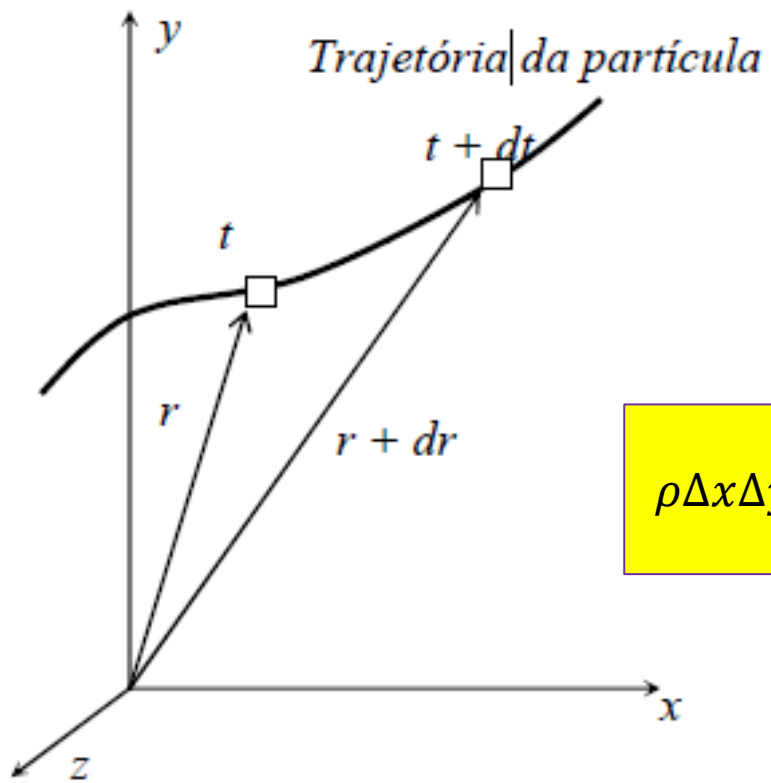


$$\Delta F_{sx} = \left(\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} \right) \Delta x \Delta y \Delta z$$

$$\Delta F_{sy} = \left(\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \tau_{zy}}{\partial z} \right) \Delta x \Delta y \Delta z$$

$$\Delta F_{sz} = \left(\frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z} \right) \Delta x \Delta y \Delta z$$

CINEMÁTICA DE UM ELEMENTO DE FLUIDO



$$\Delta \vec{F} = \Delta m \vec{a} = \rho \Delta x \Delta y \Delta z \vec{a}$$

$$\Delta \vec{F} = \Delta m \vec{a} = \rho \Delta x \Delta y \Delta z \frac{d\vec{V}}{dt}$$

$$\vec{a}_p = \frac{D\vec{V}}{Dt} = u \frac{\partial \vec{V}}{\partial x} + v \frac{\partial \vec{V}}{\partial y} + w \frac{\partial \vec{V}}{\partial z} + \frac{\partial \vec{V}}{\partial t}$$

$$\rho \Delta x \Delta y \Delta z \frac{D\vec{V}}{Dt} = \left(u \frac{\partial \vec{V}}{\partial x} + v \frac{\partial \vec{V}}{\partial y} + w \frac{\partial \vec{V}}{\partial z} + \frac{\partial \vec{V}}{\partial t} \right) \rho \Delta x \Delta y \Delta z$$

CINEMÁTICA DE UM ELEMENTO DE FLUIDO

$$\vec{F}_{total} = \vec{F}_{pressão} + \vec{F}_{gravidade} + \vec{F}_{superfície} + \dots + \vec{F}_n$$

$$\vec{F}_{pressão} = -\text{grad } P dx dy dz = -\vec{\nabla} P \Delta x \Delta y \Delta z$$

$$\vec{F}_{gravidade} = \rho dx dy dz \frac{d\vec{V}}{dt} = \rho \vec{g} * (\Delta x * \Delta y * \Delta z)$$

$$\begin{aligned} \vec{F}_{superfície} = & \left(\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} \right) \Delta x \Delta y \Delta z \\ & + \left(\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \tau_{zy}}{\partial z} \right) \Delta x \Delta y \Delta z + \left(\frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z} \right) \Delta x \Delta y \Delta z \end{aligned}$$

$$\vec{F}_{total} = \left(u \frac{\partial \vec{V}}{\partial x} + v \frac{\partial \vec{V}}{\partial y} + w \frac{\partial \vec{V}}{\partial z} + \frac{\partial \vec{V}}{\partial t} \right) \rho \Delta x \Delta y \Delta z$$

$$\vec{F}_{total} = \vec{F}_{pressão} + \vec{F}_{gravidade} + \vec{F}_{superfície} + \dots + \vec{F}_n$$

CINEMÁTICA DE UM ELEMENTO DE FLUIDO

$$\vec{F}_{total} = \vec{F}_{pressão} + \vec{F}_{gravidade} + \vec{F}_{superfície} + \dots + \vec{F}_n$$

$$\begin{aligned} & \left(u \frac{\partial \vec{V}}{\partial x} + v \frac{\partial \vec{V}}{\partial y} + w \frac{\partial \vec{V}}{\partial z} + \frac{\partial \vec{V}}{\partial t} \right) \rho \Delta x \Delta y \Delta z \\ &= -\vec{V} P \Delta x \Delta y \Delta z + \rho \vec{g} * (\Delta x * \Delta y * \Delta z) + \left(\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} \right) \Delta x \Delta y \Delta z \\ &+ \left(\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \tau_{zy}}{\partial z} \right) \Delta x \Delta y \Delta z + \left(\frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z} \right) \Delta x \Delta y \Delta z \end{aligned}$$

$$\begin{aligned} & \left(u \frac{\partial \vec{V}}{\partial x} + v \frac{\partial \vec{V}}{\partial y} + w \frac{\partial \vec{V}}{\partial z} + \frac{\partial \vec{V}}{\partial t} \right) \rho \\ &= -\vec{V} P + \rho \vec{g} + \left(\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} \right) + \left(\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \tau_{zy}}{\partial z} \right) + \left(\frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z} \right) \end{aligned}$$

CINEMÁTICA DE UM ELEMENTO DE FLUIDO

$$\begin{aligned} & \left(u \frac{\partial \vec{V}}{\partial x} + v \frac{\partial \vec{V}}{\partial y} + w \frac{\partial \vec{V}}{\partial z} + \frac{\partial \vec{V}}{\partial t} \right) \rho \\ &= -\vec{V}P + \rho \vec{g} + \left(\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} \right) + \left(\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \tau_{zy}}{\partial z} \right) \\ &+ \left(\frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z} \right) \end{aligned}$$

Separando nas componentes x,y,z

$$\left(u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} + \frac{\partial u}{\partial t} \right) \rho = -\frac{\partial P}{\partial x} + \rho \cancel{\vec{g}_x} + \left(\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} \right)$$

$$\left(u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} + \frac{\partial v}{\partial t} \right) \rho = -\frac{\partial P}{\partial y} + \rho \cancel{\vec{g}_y} + \left(\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \tau_{zy}}{\partial z} \right)$$

$$\left(u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} + \frac{\partial w}{\partial t} \right) \rho = -\frac{\partial P}{\partial z} + \rho \vec{g}_z + \left(\frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z} \right)$$

CINEMÁTICA DE UM ELEMENTO DE FLUIDO

$$\left(\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} \right) \rho = - \frac{\partial P}{\partial x} + \left(\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} \right)$$

$$\left(\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} \right) \rho = - \frac{\partial P}{\partial y} + \left(\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \tau_{zy}}{\partial z} \right)$$

$$\left(\frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} \right) \rho = - \frac{\partial P}{\partial z} + \rho \vec{g}_z + \left(\frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z} \right)$$

Considerando que as **variações** na densidade é pequena, a densidade ρ pode ser substituído em todos os lugares pela constante (referência) densidade ρ_0 , com uma grande exceção.

Esta exceção é o **termo gravitacional**, através do qual, mesmo uma pequena variação da densidade pode causar um **efeito de flutuação importante** na maioria das aplicações ambientais.

A simplificação resultante é conhecida como a **aproximação Boussinesq**.

CINEMÁTICA DE UM ELEMENTO DE FLUIDO

$$\left(\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} \right) \rho_0 = - \frac{\partial P}{\partial x} + \left(\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} \right)$$

$$\left(\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} \right) \rho_0 = - \frac{\partial P}{\partial y} + \left(\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \tau_{zy}}{\partial z} \right)$$

$$\left(\frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} \right) \rho_0 = - \frac{\partial P}{\partial z} + \rho \vec{g}_z + \left(\frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z} \right)$$

Dividindo por ρ_0

$$\left(\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} \right) = - \frac{1}{\rho_0} \frac{\partial P}{\partial x} + \frac{1}{\rho_0} \left(\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} \right)$$

$$\left(\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} \right) = - \frac{1}{\rho_0} \frac{\partial P}{\partial y} + \frac{1}{\rho_0} \left(\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \tau_{zy}}{\partial z} \right)$$

$$\left(\frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} \right) = - \frac{1}{\rho_0} \frac{\partial P}{\partial z} + \frac{\rho}{\rho_0} \vec{g}_z + \frac{1}{\rho_0} \left(\frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z} \right)$$

CINEMÁTICA DE UM ELEMENTO DE FLUIDO

$$\left(\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} \right) = -\frac{1}{\rho_0} \frac{\partial P}{\partial x} + \vec{\tau}_x$$

$$\left(\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} \right) = -\frac{1}{\rho_0} \frac{\partial P}{\partial y} + \vec{\tau}_y$$

$$\left(\frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} \right) = -\frac{1}{\rho_0} \frac{\partial P}{\partial z} + \frac{\rho}{\rho_0} \vec{g}_z + \vec{\tau}_z$$

Na Forma Vetorial

$$\frac{\partial \vec{V}}{\partial t} + (\vec{V} \cdot \nabla) \vec{V} = -\frac{1}{\rho_0} \vec{\nabla} P + \frac{\rho}{\rho_0} \vec{g} + \vec{F}_{tensão}$$

CINEMÁTICA DE UM ELEMENTO DE FLUIDO

Equações de Navier-Stokes

Para **fluidos newtonianos**, as tensões podem ser expressas em termos de **gradientes de velocidades** e propriedades dos fluidos:

Sabemos que p/ **um fluido newtoniano** a **tensão viscosa** é proporcional a taxa de deformação por cisalhamento (**taxa de deformação angular**). As tensões podem ser expressas em termos dos gradientes de velocidade e das propriedades dos fluidos como:

$$\tau_{xy} = \tau_{yx} = \mu \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right)$$

$$\tau_{yz} = \tau_{zy} = \mu \left(\frac{\partial w}{\partial y} + \frac{\partial v}{\partial z} \right)$$

$$\tau_{zx} = \tau_{xz} = \mu \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right)$$

$$\sigma_{xx} = -p - \frac{2}{3}\mu \nabla \vec{V} + 2\mu \frac{\partial u}{\partial x}$$

$$\sigma_{yy} = -p - \frac{2}{3}\mu \nabla \vec{V} + 2\mu \frac{\partial v}{\partial y}$$

$$\sigma_{zz} = -p - \frac{2}{3}\mu \nabla \vec{V} + 2\mu \frac{\partial w}{\partial z}$$

Onde p é a pressão termodinâmica local.

Equações de Navier Stokes

No caso de **fluido incompressível**, $\nabla \cdot \vec{V} = 0$, e a equação acima pode ser simplificada.

Fazendo desprezíveis as forças de campo ($B = 0$) se obtém **as Equações de Navier Stokes**.

$$\rho \frac{Du}{Dt} = -\frac{\partial p}{\partial x} + \frac{\partial}{\partial x} \left(2\mu \frac{\partial u}{\partial x} - \frac{2}{3} \mu \nabla \vec{V} \right) + \frac{\partial}{\partial y} \left[\mu \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) \right] + \frac{\partial}{\partial z} \left[\mu \left(\frac{\partial w}{\partial x} + \frac{\partial u}{\partial z} \right) \right]$$

$$\rho \frac{Dv}{Dt} = -\frac{\partial p}{\partial y} + \frac{\partial}{\partial x} \left[\mu \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) \right] + \frac{\partial}{\partial y} \left(2\mu \frac{\partial v}{\partial y} - \frac{2}{3} \mu \nabla \vec{V} \right) + \frac{\partial}{\partial z} \left[\mu \left(\frac{\partial w}{\partial y} + \frac{\partial v}{\partial z} \right) \right]$$

$$\rho \frac{Dw}{Dt} = -\frac{\partial p}{\partial z} + \frac{\partial}{\partial x} \left[\mu \left(\frac{\partial w}{\partial x} + \frac{\partial u}{\partial z} \right) \right] + \frac{\partial}{\partial y} \left[\mu \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \right) \right] + \frac{\partial}{\partial z} \left(2\mu \frac{\partial w}{\partial z} - \frac{2}{3} \mu \nabla \vec{V} \right)$$

no caso de **escoamento incompressível** permanente com viscosidade constante e incluindo as forças de campo

$$\rho \frac{Du}{Dt} = \rho g_x - \frac{\partial p}{\partial x} + \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right)$$

$$\rho \frac{Dv}{Dt} = \rho g_y - \frac{\partial p}{\partial y} + \mu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} \right)$$

$$\rho \frac{Dw}{Dt} = \rho g_z - \frac{\partial p}{\partial z} + \mu \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right)$$

Em forma vetorial pode ser representada como

$$\rho \frac{D\vec{V}}{Dt} = \rho \vec{g} - \nabla p + \mu \nabla^2 \vec{V}$$

Equações de Euler

Quando os termos viscosos são pequenos e podem ser desprezíveis ($\mu=0$) as equações resultantes são conhecidas como ***Equações de Euler***, que podem ser representadas na forma vetorial como:

$$\rho \frac{D\vec{V}}{Dt} = \rho \vec{g} - \nabla p$$